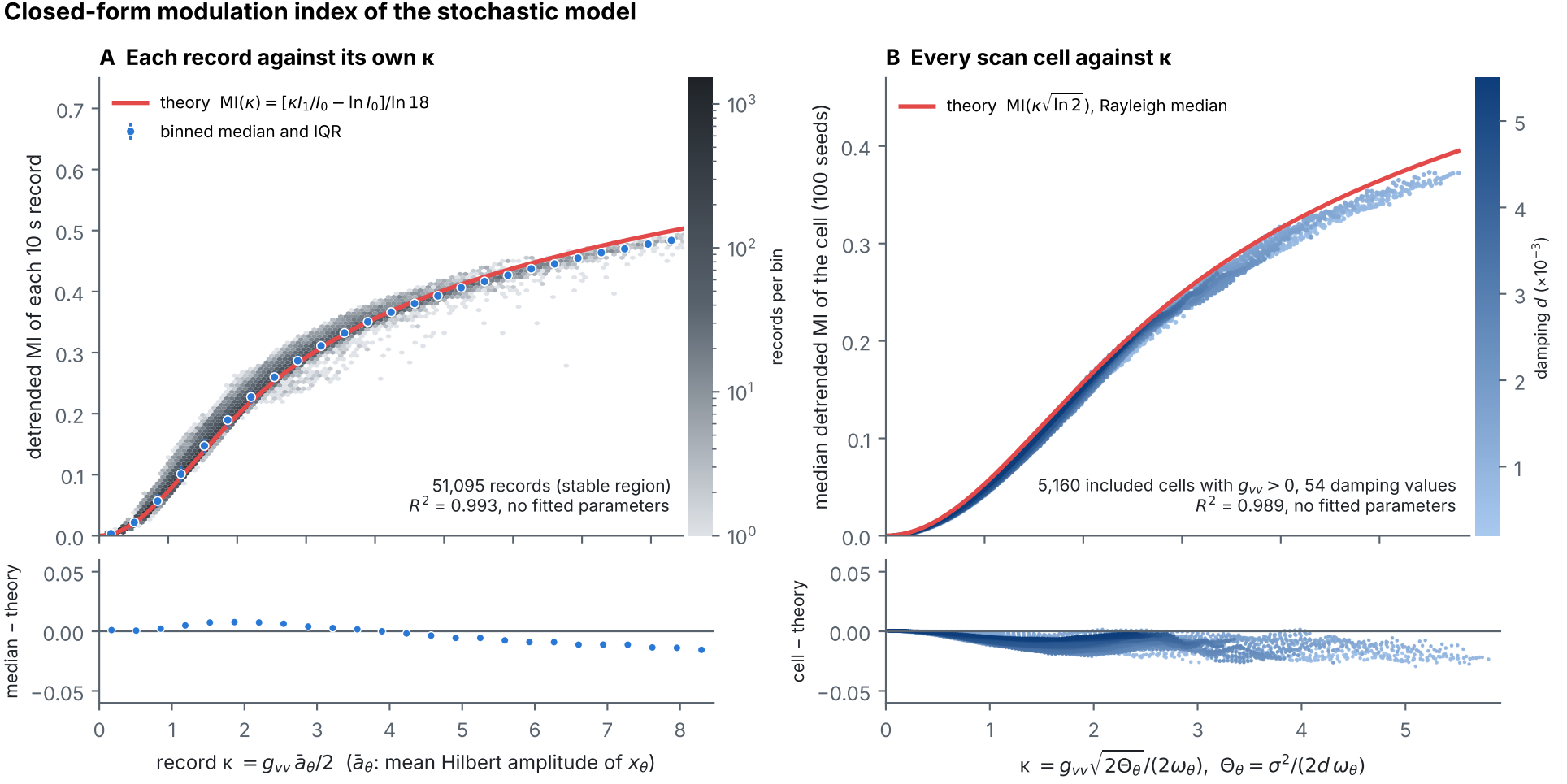

Closed-form modulation index of the stochastic model
A Each record against its own κ
theory MI(κ) = [κI1/I0 − ln I0]/ln 18
binned median and IQR
detrended MI of each 10 s record
records per bin
51,095 records (stable region)
R² = 0.993, no fitted parameters
median − theory
record κ = gvv ā_θ/2 (ā_θ: mean Hilbert amplitude of x_θ)
B Every scan cell against κ
theory MI(κ√ln 2), Rayleigh median
median detrended MI of the cell (100 seeds)
damping d (×10⁻³)
5,160 included cells with gvv > 0, 54 damping values
R² = 0.989, no fitted parameters
cell − theory
κ = gvv√(2Θ_θ)/(2ω_θ), Θ_θ = σ²/(2d ω_θ)